

Random Walk:

$$\begin{aligned} X_t &= \underline{X_{t-1}} + \underline{W_t} & (X_{t-1} &= X_{t-2} + W_{t-1}) \\ &= X_{t-2} + W_t + W_{t-1} & (X_{t-2} &= X_{t-3} + W_{t-2}) \\ &= X_{t-3} + W_t + W_{t-1} + W_{t-2} \\ &\vdots \\ &= \underline{X_0} + \underline{W_t + W_{t-1} + \dots + W_1} \end{aligned}$$

1.3. Probabilistic Descriptions of Time Series

Recall: model a time series as a sequence of RVs. $\dots, X_{t-1}, X_t, X_{t+1}, \dots$
a stochastic process

Assumption: the distribution of each X_t is continuous.

At each fixed t , then X_t is a single RV

CDF $F_t(\underline{x}) = P(X_t \leq \underline{x}) = \int_{-\infty}^x f_t(u) du$

PDF $f_t(\underline{x}) = \frac{d}{dx} F_t(\underline{x})$

Example: Random Walk

$$\begin{cases} X_t = X_{t-1} + W_t \\ X_0 = 0 \end{cases}$$

W_t : Gaussian white noise
 $\stackrel{\text{iid}}{\sim} N(0, 1)$

$X_t = W_1 + \dots + W_t$

$E[W_t] = 0$
 $\text{Var}(W_t) = 1$

Question: what is the distribution of each X_t ?

Answer: $X_t \sim N(0, t)$

Fact*: A linear combination of independent normal RVs is still a normal RV.

$\Rightarrow X_t \sim N(?, ?)$

$$E[X_t] = E[W_1 + W_2 + \dots + W_t]$$

linearity of $E[\cdot]$
 $\downarrow$
 $= E[W_1] + E[W_2] + \dots + E[W_t]$

$$= 0 + 0 + \dots + 0 = 0$$